BIOMETRIC HAIR 3D

Comprehensive Technical Architecture, System Stack, Intent, and Active Challenges

VERSION: 1.0.0 Production Handoff
DATE: November 2026
STATUS: TypeScript 0 Errors
REPO: ezung123/Biometric-Hair
TARGET: WebGL + Android / iOS

1. Executive Summary & Project Purpose

Biometric Hair 3D is an enterprise-grade, camera-first mobile and web application engineered to eliminate salon haircut anxiety. Unlike legacy 'hair makeover' tools that paste flat stickers over photos or generative AI platforms that hallucinate completely foreign faces, Biometric Hair 3D implements a **strict zero-identity-swapping mandate**. The user's genuine face, bone structure, skin texture, and expressions remain 100% untouched; only the hairstyle is computationally tailored and volumetrically seated onto their skull.

This Document: Serves as the complete technical handoff and engineering architecture audit. It provides a deep-dive analysis of the technologies employed across the pipeline, the exact mathematical principles driving hairstyle recommendations, the architectural goals we are working to achieve, and an transparent account of the active challenges and physical edge cases currently being solved.

> ***Core Mandate:*** *Zero Face Replacement. Every hairstyle is fitted directly to the user's authentic facial geometry using 468 Dense 3D cranial landmarks, the Neoclassical Tripartite Law, and 4 concentric volumetric Three.js shells.*

2. Complete Technology Stack & Architectural Layers

The system operates across three synchronized processing tiers: the client application layer, the computer vision & 3D AR WebGL engine, and the remote AI reasoning & inpainting services.

Architecture Layer	Technologies / Libraries	Technical Role & Execution Purpose

Mobile / Web Client	`React Native 0.88`
	`React 19 / Expo SDK 58`
	`Expo Router v58`
	`React Native Web`
		Unified cross-platform camera-first interface (Snapchat-style), gesture navigation, luxury dark obsidian (`#0C0A09`) & gold HUD, segmented lens carousel.

Computer Vision & Tracking	`Google MediaPipe Vision WASM`
	`468 Face Landmarks`
	`Custom Deadband Filter`
	`Temporal Crest Anchor`
		Extracts 6-DOF head pose (Pitch, Yaw, Roll, Sway X, Depth Z) at 30 FPS. Tracks natural Trichion hairline and frontotemporal skull crests without re-render cascades.

3D Volumetric Engine	`Three.js WebGL (r187)`
	`4-Layer Concentric Shells`
	`Sobel 3D Normal Mapping`
	`3-Point Specular Studio Lighting`
		Constructs true volumetric depth (+0mm, +8mm, +16mm, +25mm) with micro-strand relief, dynamic inertia sway physics, and per-hairstyle skull seating.

AI Reasoning & Inpainting	`Meta Muse Spark Open (Baseten)`
	`Meta SAM (Modal Serverless)`
	`Photorealistic Synthesis Canvas`
		Muse analyzes Neoclassical facial thirds and generates styling prescriptions. Modal SAM and local canvas inpainter suppress old silhouette and cast ambient forehead shadows.

Native Hardening	`Expo Prebuild Engine`
	`Android Camera/Media Perms`
	`Google Play Manifest`
		Production Android package `com.ezung.biometrichair`, versionCode 1, scoped media permissions (`READ_MEDIA_IMAGES`), and cold-boot theme persistence.

3. Cephalometric Mathematics & The Golden Ratio Engine

Rather than relying on arbitrary fashion tags, Biometric Hair 3D matches haircuts based on proven craniofacial anthropometry and the **Golden Ratio phi (1.618)**:

- **Neoclassical Tripartite Facial Thirds:** The face is segmented into Upper Third (Trichion hairline → Nasion brow), Mid Third (Nasion → Subnasale nose base), and Lower Third (Subnasale → Gnathion chin tip). A canonical balanced face possesses 33.3% / 33.3% / 33.3% distribution. The AI calculates third variance to prescribe specific crown volume or forehead fringe coverage.
- **Facial Index (Morphological Height-to-Width Ratio):** Measures anatomical face height divided by Bizygomatic cheekbone width × 100. Categorizes morphology strictly into *Oblong (>90)*, *Oval (85–90)*, *Round (<80)*, or *Square (angular jawline)*.
- **Cephalometric Calibration Matrix:** Each hairstyle maintains precise mathematical seating constraints defined in STYLE_CALIBRATION:

Hairstyle ID	Category	Gender	Hairline (yOffset)	Scale Factor	Cranial Wrap (Z)	Flank Drape
`curtain-fringe`	Fringe	Men	0.22	1.06	74 mm	No
`textured-crop`	Crop	Men	0.16	1.02	68 mm	No
`classic-quiff`	Quiff	Men	0.18	1.04	72 mm	No
`taper-fade`	Fade	Men	0.12	1.02	68 mm	No
`buzz-cut`	Fade	Men	0.10	0.98	62 mm	No
`afro-taper`	Fade	Men	0.22	1.10	80 mm	No
`curly-fringe`	Curls	Men	0.18	1.06	74 mm	No
`layered-waves`	Waves	Women	0.24	1.08	78 mm	No
`french-bob`	Bob	Women	0.14	1.04	70 mm	No
`sleek-pixie`	Pixie	Women	0.16	1.02	68 mm	No
`box-braids`	Braids	Women	0.14	1.08	84 mm	Yes

4. Project Objectives & What We Are Trying To Achieve

The overarching goal is to deploy the world's most physically accurate and visually compelling hair makeover application on both the Google Play Store and Apple App Store, offering a zero-friction user experience:

Objective	Target Milestone Metric	Engineering Delivery Strategy
Snapchat/Lensa AR Parity	`60 FPS render loop` `<16ms frame budget`	Direct GPU WebGL rendering, deadband landmark filtering, and eliminating React state cascades during steady pose tracking.
Snug Cephalometric Fit	`<2mm hairline variance` `Zero helmet bloating`	Recalibrated projection scales from 1.45 down to 0.35–0.45, utilizing temporal crest width and Trichion hairline bounds.
Ethnic & Texture Inclusivity	`100% hair type coverage` `Type 1A straight to 4C coils`	Dedicated volumetric calibration profiles for Afro Taper (dense 4C coils), Curly Fringe (3A-3C spirals), and Box Braids (long draping plaits).
Luxury Non-AI Aesthetic	`Zero emoji clutter` `Cream/Obsidian UI`	Sleek iOS-inspired typography, real-time gold holographic telemetry scanner, and buttery smooth drawer interactions.
App Store Deployment	`Google Play & App Store compliant`	Production Android package `com.ezung.biometrichair`, versionCode 1, scoped media permissions, and zero TS errors.

5. Critical Engineering Challenges & What We Are Struggling With

While the 3D pipeline and hairstyle seating have made massive leaps, building a real-time biometric AR engine across web and mobile introduces five fundamental physics and rendering bottlenecks:

1. WebGL Transparency Sorting & Strand Alpha Bleed Across 4 Shells

When rendering 4 concentric volumetric shells (+0mm, +8mm, +16mm, +25mm) with semi-transparent alpha textures, standard WebGL depth-buffer writes (depthWrite: true) cause background shells to be clipped by the bounding boxes of foreground shells. Conversely, disabling depthWrite (depthWrite: false) causes alpha sorting anomalies when the user turns their head rapidly—strands on the far side of the head occasionally render in front of foreground locks. **Current Remediation:** Fine-tuning alphaTest (0.04–0.06 threshold) and dual-pass render queues to enforce strict front-to-back sorting.

2. Consumer Webcam Lighting vs Studio Normal Map Shaders

The Three.js scene employs studio three-point directional lighting (Key, Gold Rim, Cool Fill) and real-time Sobel normal maps to produce rich strand specular sheen. However, end-user webcams and smartphone front cameras often operate under harsh monochromatic fluorescent ceiling lights or dim bedroom settings. This creates a lighting mismatch where the augmented hair appears brightly studio-lit against a dark or grainy user photo. **Current Remediation:** Sampling cheek/forehead skin luminance in real-time to modulate Three.js ambient light intensity dynamically.

3. Monocular 2D Cranial Depth Extrapolation on Pitch / Yaw

Standard RGB cameras provide only 2D pixel coordinates. MediaPipe infers pseudo-3D landmarks, but deep pitch (tilting chin up or down) and yaw (turning left or right beyond 35 degrees) causes perspective distortion. Because human hair extends far behind the visible forehead line, yaw rotations can cause the posterior edge of the hair mesh to disconnect from the skull contour. **Current Remediation:** Anchoring the rotation origin at the frontotemporal crests rather than the nose tip, and clamping rotation gains.

4. WebGL Context Contention Between MediaPipe and Three.js

Both Google MediaPipe Face Landmarker (WASM/WebGL pipeline) and the Three.js 3D hair canvas require active WebGL contexts. On low-end Android devices or laptops with integrated GPUs, running two concurrent WebGL contexts alongside 1024x1024 video textures can cause GPU memory pressure, thermal throttling, or occasional context loss events. **Current Remediation:** Throttle MediaPipe inference from 60 Hz to 24–30 Hz with deadband filtering while keeping Three.js rendering at 60 FPS.

5. Native Android Camera Stream Bridging (Expo vs Vision Camera)

On web browsers, navigator.mediaDevices.getUserMedia directly binds a zero-copy HTML5 <video> element to both MediaPipe and Three.js. On native Android and iOS, raw camera frame buffers cannot be shared directly with web-based WebGL without custom frame processors or GLView native surface textures. Passing high-resolution camera frames across the React Native JavaScript bridge incurs significant serialization latency. **Current Remediation:** Implementing platform-specific branching with native frame-processor bindings for Android production.

6. Immediate Roadmap & Actionable Next Steps

- **[Phase 1 Complete]** Eliminated white occluder cutout hole, calibrated snug Three.js projection (0.35–0.45), built 4 volumetric shells.
- **[Phase 2 Complete]** Replaced broken split-slider with luxury Biometric AI Scanning HUD; guaranteed hairstyle try-on rendering.
- **[Phase 3 Complete]** MediaPipe deadband jitter suppression, temporal crest anchoring, Android Play Store permissions in app.json.
- **[Phase 4 Complete]** Calibrated ethnic & curly profiles in STYLE_CALIBRATION (Afro Taper, Curly Fringe, Box Braids) with 1024x1024 PNGs.
- **[Upcoming Priority 1]** Deploy Ambient Luminance Normalization to match hair specular tone with user room lighting.
- **[Upcoming Priority 2]** Implement native Android VisionCamera / GLView frame processor bridge for standalone APK build.
- **[Upcoming Priority 3]** Submit release bundle to Google Play Internal Testing Track for closed community validation.